MALLIKARJUN NEERALAGI

+91-6363538436 | neeralagimallikarjun@gmail.com | linkedin.com/in/mallikarjun-neeralagi | github.com/mallikarjun-neeralagi

PROFILE SUMMARY

Electronics and Communication Engineering student with hands-on experience in **Embedded Systems, Embedded C, PCB Design, and IoT Development**. Proficient in **C, C++, Embedded C, ESP32, STM32, Raspberry Pi, UART, SPI, and I²C**, with practical exposure to **Altium Designer, KiCad, SMT assembly, hardware testing, debugging, and system integration**. Interested in embedded firmware, product engineering, and hardware-software integration.

TECHNICAL SKILLS

Programming	C, C++, Embedded C, Python
Embedded Systems	STM32, ESP32, Arduino, Raspberry Pi, GPIO, Timers, PWM, ADC, Interrupts
Communication Protocols	UART, SPI, I²C, CAN
Data Structures	Arrays, Strings, Linked Lists, Stacks, Queues, Searching, Sorting, Bit Manipulation
RTOS	FreeRTOS – Tasks, Scheduling, Queues, Semaphores, Mutexes, Software Timers
Linux	Linux Fundamentals, Shell Commands, Processes, Threads, File System
PCB Design	Altium Designer, KiCad
Tools	STM32CubeIDE, Arduino IDE, Git, Matplotlib
Hardware	Sensors, Displays, SMT Assembly, Pick-and-Place Machines, PCB Inspection, Hardware Testing and Debugging

PROJECTS

ThermaGuard AI – Thermal Object Detection

- Developed ESP32 firmware in **Embedded C** to interface the MLX90640 thermal infrared array sensor through **I²C**.
- Acquired and transmitted thermal sensor data through serial communication for real-time monitoring.
- Developed a Python-based thermal heatmap for visualization and analysis of temperature patterns.
- Integrated embedded hardware and PC-side processing for real-time thermal object detection.

STM32-Based Embedded System with Peripheral Interfacing

- Developed embedded firmware in **C/Embedded C** for STM32-based sensor and peripheral control.
- Implemented GPIO, timers, PWM, ADC and interrupt-based event handling for real-time hardware control.
- Interfaced sensors and peripherals using **UART, SPI and I²C** communication protocols.
- Performed firmware debugging and hardware communication testing through serial monitoring and systematic fault isolation.

IoT-Based Home Automation System

- Developed an ESP32-based home automation system using relay modules and Wi-Fi communication for remote appliance control.
- Implemented microcontroller-based appliance control and automated scheduling for improved energy efficiency.

INDUSTRIAL TRAINING

Embedded Systems and PCB Design Trainee — **Vithsutra Technologies Pvt. Ltd.**

- Designed and modified PCB layouts using **Altium Designer and KiCad** for embedded hardware projects.
- Observed and applied industry-standard PCB assembly and quality assurance workflows.
- Gained hands-on experience in **SMT manufacturing** and Pick-and-Place machine operation.
- Assisted in hardware testing, debugging, and system integration.

EDUCATION

Alva's Institute of Engineering and Technology, Moodbidri — 2023 – 2027
Bachelor of Engineering in Electronics and Communication Engineering — **CGPA: 8.4**

Prerana PU College, Belagavi — 2021 – 2023
Pre-University (PCMC) — **Percentage: 86%**

SRMRS, Khanapur — 2021
SSLC — **Percentage: 93%**

CERTIFICATIONS

- NPTEL – Embedded Systems
- Scalar – Python Programming
- Coursera – Altium Design, C, C++, Embedded C